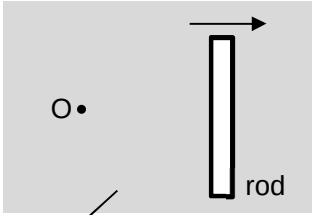
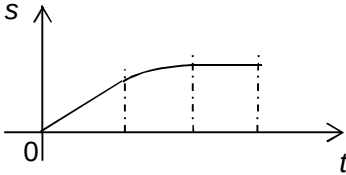
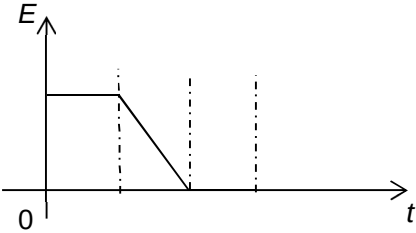
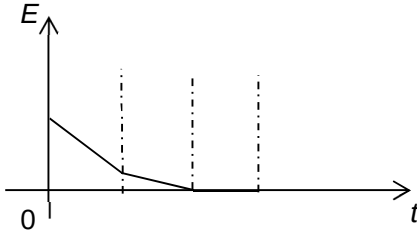
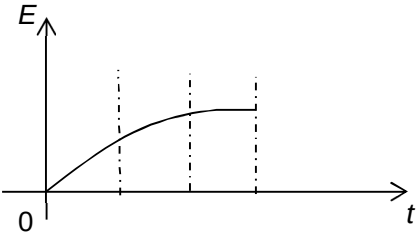
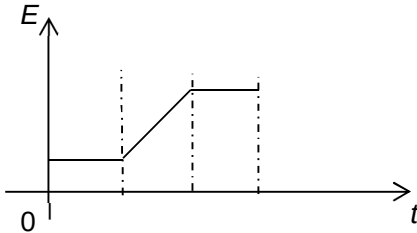


25

Diagram A shows an aluminium rod moving away from point O, at right angles to a uniform magnetic field. Diagram B shows the variation with time t of its displacement s from O.

	 region of magnetic field Diagram A  Diagram B Which graph best shows the variation with t of the e.m.f. E induced in the rod?
	A  B 
	C  D 

Answer: A

By Faraday's law, e.m.f. induced is the rate of change of magnetic flux linkage. Since the rate of change of magnetic flux linkage is proportional to the velocity of the rod, the e.m.f.-time graph has the same shape as the velocity-time graph. i.e. refers to the gradient of s-t graph.